IND-ACC-65 Failure Response - Arista Access Node Routing Protocol Flap SOP

Node: IND-ACC-65 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1385

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on the Arista Access Node IND-ACC-65 located at the Indiranagar Exchange. The failure is attributed to the dynamic nature of the routing protocols and the hardware capabilities of the Arista node. BGP route churn occurs due to the continuous exchange of routing information between neighbor routers, while OSPF adjacency loss is caused by temporary network partitions or interface errors. ISIS LSP flooding is triggered by the node's attempt to re-establish routing information in response to changes in the network topology. This failure domain is specific to the Access role on Arista equipment due to the reliance on routing protocols to manage and forward traffic.

Pre-requisite Checks

- check: Verify the power and network connectivity to the Indiranagar Exchange
- check: Confirm the redundant link status on the Arista Access Node IND-ACC-65

Potential Root Causes

Cause	Probability
BGP Route Churn	Medium
OSPF Adjacency Loss	High
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version 4, Description: This command displays the BGP table version, which can help identify any recent updates or churn in the routing information.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command shows the OSPF neighbor status, which can help identify any adjacency loss or interface errors.

Command: show isis database, Expected Output: LSP Seq/LSA Seq: 0x12345678/0x90123456, Description: This command displays the ISIS LSP database, which can help identify any flooding or re-establishment of routing information.
```

Resolution Steps

Step 1: Step: Restart the BGP process, Command: process restart bgp, Description: This command restarts the BGP process, which can help resolve BGP route churn and re-establish routing information.

Step 2: Step: Clear the OSPF neighbor table, Command: clear ospf neighbor, Description: This command clears the OSPF neighbor table, which can help resolve OSPF adjacency loss and re-establish OSPF routing information.

Step 3: Step: Clear the ISIS LSP database, Command: clear isis database, Description: This command clears the ISIS LSP database, which can help resolve ISIS LSP flooding and re-establish ISIS routing information.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version 5, Description: This command verifies that the BGP table version has increased, indicating that the BGP process has been restarted successfully.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command verifies that the OSPF neighbor status has been restored, indicating that the OSPF adjacency loss has been resolved.

Command: show isis database, Expected Output: LSP Seq/LSA Seq: 0x12345678/0x90123456, Description: This command verifies that the ISIS LSP database has been cleared, indicating that the ISIS LSP flooding has been resolved.
```

Roll-back Procedures

- Step: Restore the BGP process, Command: process restart bgp, Description: This command restores the BGP process to its previous state, which can help roll back any changes made during the troubleshooting process.

- Step: Restore the OSPF neighbor table, Command: clear ospf neighbor, Description: This command restores the OSPF neighbor table to its previous state, which can help roll back any changes made during the troubleshooting process.

- Step: Restore the ISIS LSP database, Command: clear isis database, Description: This command restores the ISIS LSP database to its previous state, which can help roll back any changes made during the troubleshooting process.